

VIDITH BALASA

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EDUCATION

MS in Artificial Intelligence, University of San Francisco

Jun 2023 - Jun 2024

BA in Philosophy, University of California, Santa Cruz

Sep 2018 - Jun 2022

EXPERIENCE

Siemens

Nov 2024 - Present

AI Engineer

Cincinnati, OH

- Led development of the embedding pipeline making 400+ TB of ASML engineering data searchable, onboarding over 20,000 users onto Teamcenter AI
- Built a multi-agent system in production that lets users control Teamcenter through natural language, using LangGraph and an MCP server that exposes product operations as tools
- Provisioned and configured AI infrastructure across AWS and Azure using Terraform and Ansible, enabling migration from on-premises to cloud deployment
- Built an automated evaluation pipeline in GitLab CI with Langfuse tracing, translating agent performance into metrics reviewed by non-technical stakeholders
- Extended Teamcenter's core C++ services behind a REST API layer in a large legacy SOA codebase, and shipped React frontend features for AI capabilities in the product
- Researched and shipped an image-to-CAD retrieval pipeline with the Siemens NX team, letting engineers photograph a physical part and retrieve its exact 3D model and part ID by similarity search over hundreds of automated 2D renders per CAD model, reaching 85% top-k accuracy on a 5,000-part catalog

Square

Oct 2023 - Oct 2024

Machine Learning Engineer – Intern

San Francisco, CA

- Built binary classifiers in Python predicting which prospective merchants would convert from Square's direct mail campaigns, training XGBoost, decision trees, and neural networks on hundreds of GBs of data
- Improved on the production model baseline through feature engineering and hyperparameter tuning, raising F1 by 5% on the deployed test set
- Developed customer lifetime value models for the growth team, building training datasets from Snowflake with SQL and orchestrating training and scoring pipelines in Prefect

Siemens

May 2023 - Jul 2023

Software Engineer – Intern

Remote

- Integrated LLM-powered question answering over customer document repositories into Teamcenter, working within the platform's legacy Java codebase
- Built a natural-language-to-SQL pipeline scoped to Teamcenter database schemas, letting non-technical users query engineering data conversationally

Diversity

Jun 2022 - Sep 2022

Software Engineer – Intern

Remote

- Built and deployed image classification models on AWS powering a diversity-focused recruiting product that surfaced underrepresented candidates to employer ATS systems
- Developed data pipelines collecting and processing LinkedIn profile data for model training and evaluation

PERSONAL ENDEAVORS

ScaleAI Hackathon Winner One of three top prize winners at the annual Scale AI hackathon. Built an LLM molecular generation model fine-tuned on a SMILES dataset to accelerate drug discovery for chemical engineers.

Fog City Rocketry Officer for fog city rocketry club. Built a 3D printed rocket with a custom H-class solid fuel engine that reached 10,000 feet. Worked on IMU and flight computer algorithms.